

Bird Abundance

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Set up

Packages

```
# general use  
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr      1.1.4      v readr      2.1.5  
v forcats    1.0.0      v stringr    1.5.1  
v ggplot2    3.5.2      v tibble     3.2.1  
v lubridate  1.9.4      v tidyr      1.3.1  
v purrr      1.0.4  
-- Conflicts ----- tidyverse_conflicts() --  
x dplyr::filter() masks stats::filter()  
x dplyr::lag()     masks stats::lag()  
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become
```

```
library(here)
```

here() starts at /Users/ethancastelazo/GitHub/final_193DD_proj

```
library(janitor)
```

Attaching package: 'janitor'

The following objects are masked from 'package:stats':

chisq.test, fisher.test

```
library(snakecase)

#Wrap Axis Label
library(scales)
```

Attaching package: 'scales'

The following object is masked from 'package:purrr':

discard

The following object is masked from 'package:readr':

col_factor

```
#reading .xlsx files
library(readxl)

# visualize missing data
library(naniar)

# visualizing
library(patchwork)
```

Data

```
birds <- read_csv(here("data", "birds.csv"))
```

Rows: 15377 Columns: 6

-- Column specification -----

Delimiter: ","

chr (5): observation_date, season, species, repeat_observation, e_bird_group

dbl (1): count

i Use ``spec()`` to retrieve the full column specification for this data.
i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

```
water <- read_csv(here("data", "venoco-water.csv"))
```

Rows: 31974 Columns: 7

-- Column specification -----

Delimiter: ","

dbl (5): comp_level_ft, temperature, year, month, wy

date (1): date

time (1): time

i Use ``spec()`` to retrieve the full column specification for this data.
i Specify the column types or set ``show_col_types = FALSE`` to quiet this message.

counting birds

```
number_of_species <- birds %>%  
  mutate(n_species = n_distinct(species)) %>%  
  mutate(n_groups = n_distinct(e_bird_group))
```